

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is:

One potential explanation for the website's connection timeout error message is: an automated alert from the monitoring system indicated a problem with the web server. I attempt to visit the company's website, but received a connection timeout error message in your browser.

The logs show that:

A large number of TCP SYN requests coming from an unfamiliar IP address. The web server appears to be overwhelmed by the volume of incoming traffic and is losing its ability to respond to the abnormally large number of SYN requests. This event could be: You suspect the server is under attack by a malicious actor.

This event could be:

This event could be a single-source SYN flood denial-of-service (DoS) attack.

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:

1. SYN – The client sends a synchronize packet to start the connection.

2. SYN-ACK – The server acknowledges and responds to the request.

3. ACK – The client confirms, completing the handshake.

Explain what happens when a malicious actor sends a large number of SYN packets all at once:

When a malicious actor sends a large number of SYN packets all at once, the server receives thousands of connection requests but never gets the final ACKs. The server holds these half-open connections, using up memory and processing power until it becomes unable to respond to legitimate users, causing connection timeouts.

Explain what the logs indicate and how that affects the server:

The logs indicate a large number of SYN requests from one unfamiliar IP address. This shows the server is being overwhelmed by incomplete handshakes, resulting in a single-source SYN flood DoS attack that makes the website unavailable.

Additional comments

The IT team became aware of the incident when an attempt to access the company's website resulted in a connection timeout. After checking the monitoring dashboard, a critical alert indicated that the web server was not responding. The issue was immediately reported to the IT team, who escalated it to the security team for further investigation. When the security team reviewed the server logs, they observed a large number of SYN packets coming from a single unfamiliar IP address. Based on this evidence, they concluded that the server was likely experiencing a single-source SYN flood attack.

To mitigate the attack, the team adjusted the web server configuration to rate-limit incoming SYN packets from the malicious IP address. This reduced the impact of the attack by slowing the attacker's connection attempts, allowing legitimate users to regain access even though some disruption continued. After stabilizing the server, the team documented the incident in detail. This documentation provided evidence for post-incident review, helped improve future detection and response processes, and ensured the organisation could learn from the attack.